

DOCUMENT 3: TECHNOLOGY STACK

S.NO	ARCHITECTURE COMPONENT	TECHNOLOGY CHOSEN	JUSTIFICATION / PURPOSE
1	Frontend / Client-Side	HTML5, CSS3, Vanilla JavaScript, Jinja2 Templates	Fast rendering with zero build tools needed. Jinja2 integrates natively with FastAPI for server-side template rendering. Dark glassmorphism design with responsive breakpoints. No React/Angular overhead needed for this scope
2	Backend / Server-Side	Python FastAPI + Uvicorn	High-performance async framework with automatic OpenAPI/Swagger docs generation. Pydantic for input validation. Lightweight, excellent for AI/ML API integrations. Async support for concurrent requests
3	Database / Data Storage	File System (static/panels/, static/exports/)	No database required for MVP. Generated panel images stored as PNGs. PDFs exported on-demand and served statically. Simple, fast, no DB setup needed
4	Cloud / Hosting / Deployment	Localhost (Uvicorn) + Future: Render/Railway/Hugging Face Spaces	Easy local development with hot-reload. Future cloud deployment options include Render (free tier with auto-scaling), Railway, or Hugging Face Spaces for AI apps
5	Version Control & CI/CD	Git + GitHub	Industry-standard version control enabling collaboration, code review via PRs, issue tracking, and project management. README-driven documentation
6	Third-Party APIs / Tools	Google Gemini 1.5 Flash (outline), Gemini 1.5 Pro (story/narration), Hugging Face Diffusers (image gen with Stable Diffusion/FLUX.1-schnell), FPDF2 (PDF export), Pillow (image processing)	Gemini: best cost-quality for structured text. HF Diffusers: open-source SOTA image generation. FPDF2: lightweight dependency-free PDF with Unicode support